Physical Chemistry Cume April 4, 2003

Microwave (MW) spectroscopy is one of the two main methods for determining the structures of gas-phase molecules.

(A) Describe the operating principle of MW spectroscopy, sketch the instrument, and give its frequency range. (20 p)

(B) Molecular states are characterized by electronic, vibrational, rotational and nuclear spin quantum numbers. The changes of which of these are measured primarily in MW spectroscopy? (10 p)

(C) The equilibrium C-O distance in carbon monoxide is 1.128 Å, the atomic masses are 12.000 amu and 15.9949 amu (1 amu = 1.66054E-27 kg). Calculate the moment of inertia I of CO, and the frequencies of the J=0 → 1 and J=1 → 2 transitions in SI units. Help: The effective mass μ is given by $\mu=m_1m_2/(m_1+m_2)$, and the energy levels are given by $E(J)=(\hbar^2/2I)J(J+1)$. The value of the Planck constant is $\hbar$=1.054573E-34 kgm^2s^{-1}; h=6.626075E-34 kgm^2s^{-1}. (40 p)

(D) Which of the following molecules does **not** have a microwave spectrum:

(a) H_2 (b) BF_3 (c) N_2O (Note: NNO) (d) CH_3Cl (e) NH_3 (15 p)

(E) If a molecule does not have a MW spectrum, the same information yielded by MW spectroscopy (essentially the moments of inertia) can still be determined experimentally from other spectroscopic observations. How? (15 p)